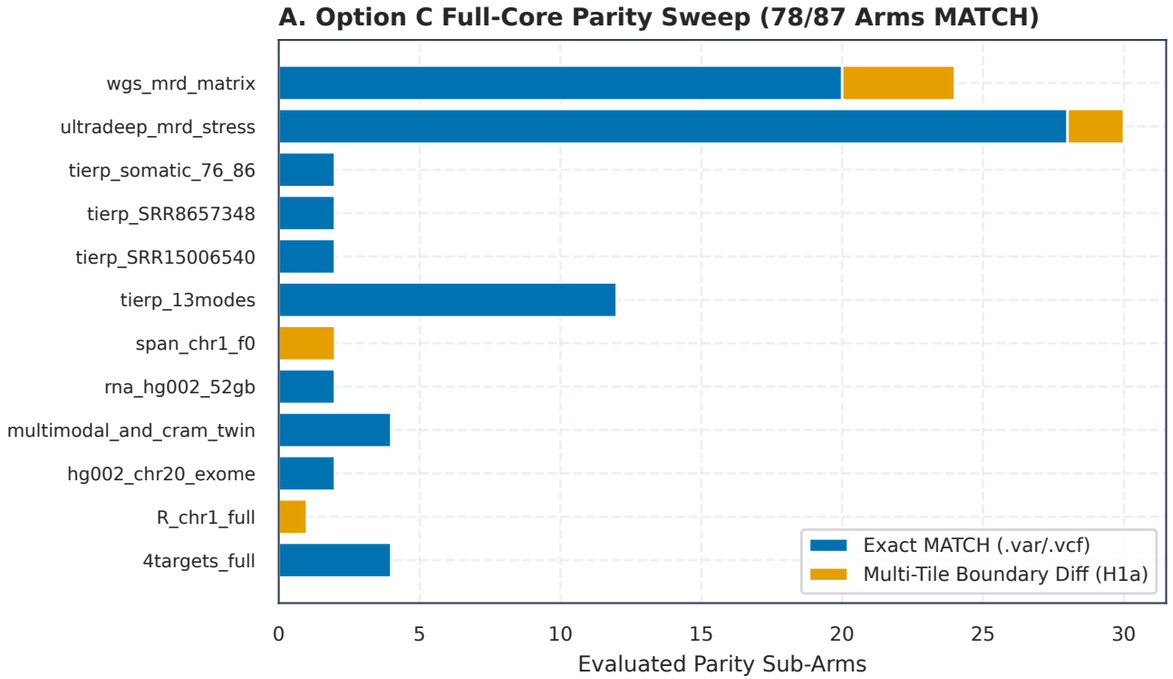
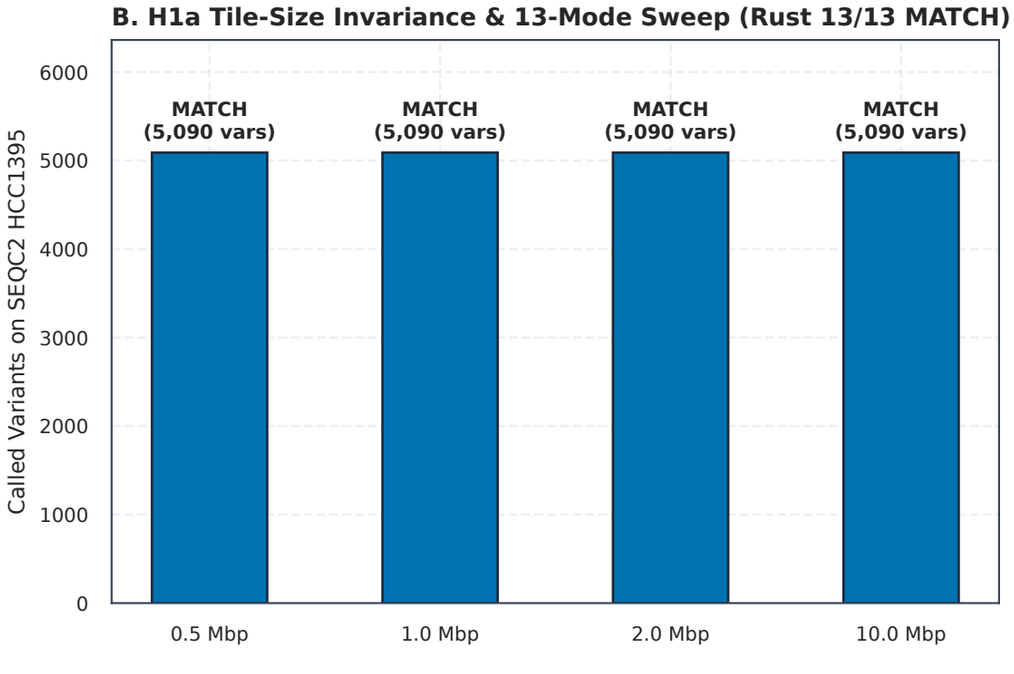
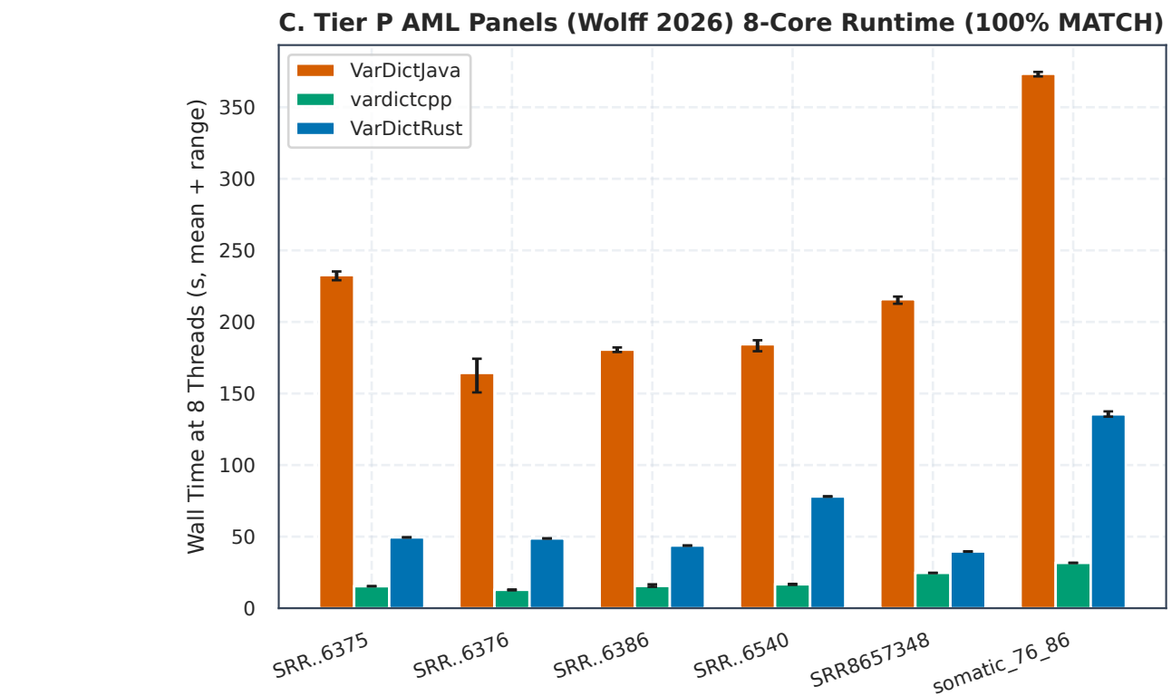
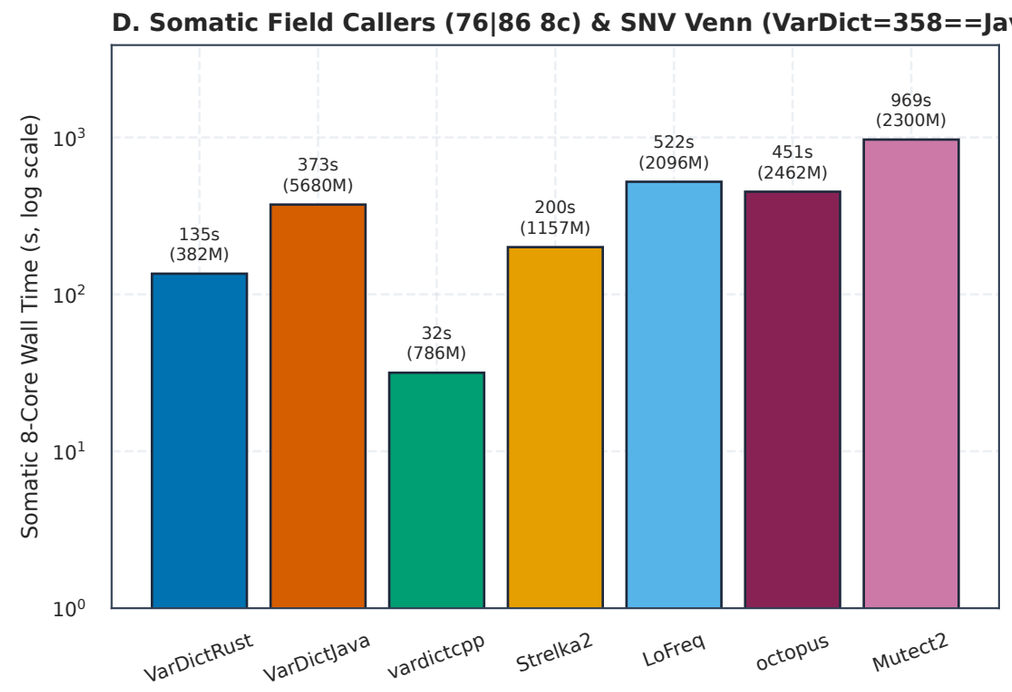

A. Option C Full-Core Parity Sweep (78/87 Arms MATCH)
wgs_mrd_matrix
ultradeep_mrd_stress
tierp_somatic_76_86
tierp_SRR8657348
tierp_SRR15006540
tierp_13modes
span_chr1_f0
rna_hg002_52gb
multimodal_and_cram_twin
hg002_chr20_exome
R_chr1_full
4targets_full
Exact MATCH (.var/.vcf)
Multi-Tile Boundary Diff (H1a)
0
5
10
15
20
25
30
Evaluated Parity Sub-Arms


B. H1a Tile-Size Invariance & 13-Mode Sweep (Rust 13/13 MATCH)
Called Variants on SEQC2 HCC1395
0
1000
2000
3000
4000
5000
6000
MATCH
(5,090 vars)
MATCH
(5,090 vars)
MATCH
(5,090 vars)
MATCH
(5,090 vars)
0.5 Mbp
1.0 Mbp
2.0 Mbp
10.0 Mbp


C. Tier P AML Panels (Wolff 2026) 8-Core Runtime (100% MATCH)
VarDictJava
vardictcpp
VarDictRust
Wall Time at 8 Threads (s, mean + range)
0
50
100
150
200
250
300
350
SRR..6375
SRR..6376
SRR..6386
SRR..6540
SRR8657348
somatic_76_86


D. Somatic Field Callers (76|86 8c) & SNV Venn (VarDict=358==Java)
Somatic 8-Core Wall Time (s, log scale)
$10^0$
$10^1$
$10^2$
$10^3$
135s
(382M)
373s
(5680M)
32s
(786M)
200s
(1157M)
522s
(2096M)
451s
(2462M)
969s
(2300M)
VarDictRust
VarDictJava
vardictcpp
Strelka2
LoFreq
octopus
Mutect2